

**Math 2551 Worksheet: Quadric Surfaces**

1. For the following, identify and describe (for example, which way is it oriented? what are the cross-sections?) the type of surface.

(a)  $x^2 + 4z^2 = 16$ .

(b)  $9x^2 + z^2 + y^2 = 9$ .

(c)  $y^2 = 3x^2 + 3z^2$ .

(d)  $z = x^2 + y^2 + 4$ .

(e)  $x^2 + y^2 = 16 - z^2$